The documentation and process conversion measures necessary to comply with this revision shall be completed by 12 September 1997

METRIC

MIL-PRF-19500/629A
12 June 1997
SUPERSEDING
MIL-S-19500/629
14 February 1995

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, RECTIFIER,
SCHOTTKY BARRIER, TYPES 1N6702 AND 1N6702US
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500, and two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figures 1, 2, and 3 (JANC die) dimensions.

1.3 Maximum ratings.

Types	V_{RWM} 1/	I_{O1} 2/	I_{FSM}	T_{STG}	T_J
	V (pk)	A dc	A (pk)	°C	°C
1N6702, 1N6702US	40	5.0	150	-65 to +150	-65 to +125

1/ Derate linearly at 1.2 V/ °C above T_L or T_{EC} = +90 °C where T_L is at L = 9.52 mm (.375 inch).
2/ Derate linearly at 71 mA/ °C above T_L or T_{EC} = +55 °C where T_L is at L = 9.52 mm (.375 inch).

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25 °C.

Types	Max V_{FM1} I_{FM} = 3.0 A	Max V_{FM2} I_{FM} = 5.0 A	Max V_{FM3} I_{FM} = 15.7 A	Max I_{RM} V_{RM} = 40 V dc pulsed method (see 4.5.1)		Max $R_{\theta JL}$ or $R_{\theta JEC}$ 9.52 mm (.375 inch) lead length or end cap	Max $Z_{\theta JX}$
				T_J = +25 °C I_{RM1}	T_J = +100 °C I_{RM2}		
	V (pk)	V (pk)	V (pk)	mA	mA	°C/W	°C/W
1N6702	0.44	0.47	0.60	0.20	25.0	20	2.5
1N6702US	0.44	0.47	0.60	0.20	25.0	10	2.5

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated detail specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

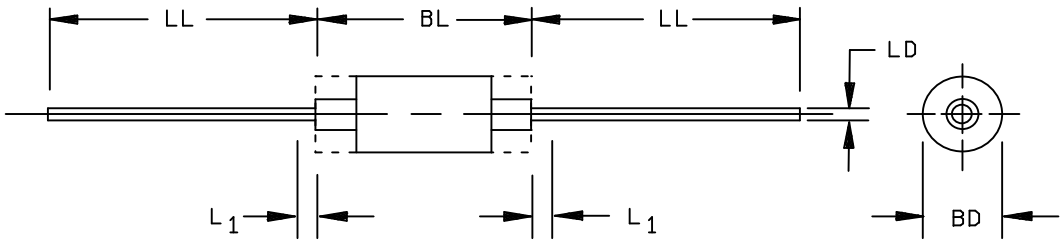
3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Design, construction, and physical dimensions. The design, construction, and physical dimensions for the purpose of interchangeability shall be as specified on figures 1, 2, and 3 herein. Plastic packages are prohibited. The US Government's preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.4.1 Lead material and finish. Lead material shall be copper clad steel with a minimum of 70 percent copper by weight. Lead finish shall be in accordance with MIL-PRF-19500 and MIL-STD-750. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be constructed in a manner and using materials which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

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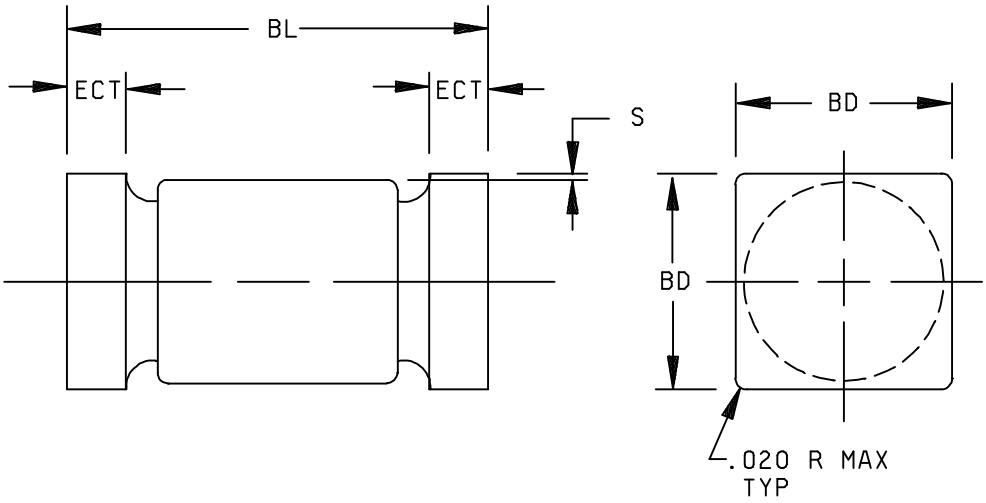


Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
LD	1.19	1.37	.047	.054	
BD	3.68	4.70	.145	.185	3
BL	3.81	7,62	.150	.300	
LL	25.40	33.02	1.000	1.300	
L_1		1.27		.050	4

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. Symbol BD shall be measured at the largest diameter.
4. Lead diameter is not controlled in this zone to allow for flash, lead finish build-up, and mirror irregularities other than heat slugs.

FIGURE 1. Physical dimensions of 1N5822.

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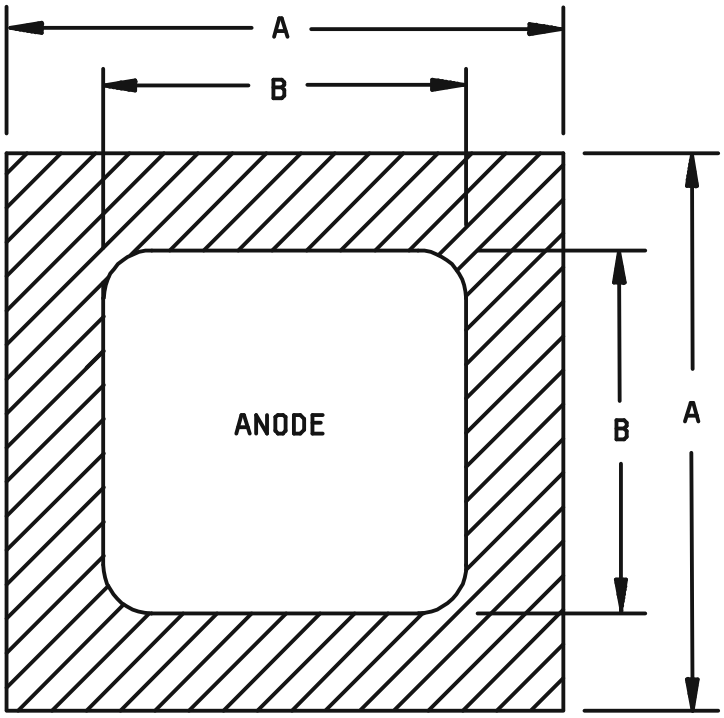


Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
BL	5.20	6.22	.205	.245
BD	4.65	5.13	.183	.202
ECT	0.48	0.71	.019	.028
S	0.08	- - -	.003	- - -

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. In accordance with ANSI Y14.5M, diameters are equivalent to ∅X symbology.

FIGURE 2. Physical dimensions of surface mount family, (D-5C).

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Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	2.24	2.29	.088	.090
B	1.98	2.03	.078	.080

Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness. . . . . 25,000 Å minimum.
Gold thickness . . . . . 4,000 Å minimum.
Chip thickness . . . . . 0.254 mm (10 mils) ±.051 (2 mils).

FIGURE 3. JANC (A-version) die dimensions.

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3.4.2.1 Surface mount (US). The surface mount (US) version shall be considered structurally identical to the non surface mount (US) version except for lead attach.

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.5.1 Marking for surface mount (US) devices. Surface mount (US) suffix parts are to be marked with the polarity identification. Initial container package marking will be in accordance with MIL-PRF-19500.

3.6 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.2.2 JANHC and JANKC die devices. Qualification for these devices shall be in accordance with MIL-PRF-19500. This testing may be performed on a TO-5 package in lieu of the axial leaded packag

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4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-PRF-19500 (appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500	Measurement		
	JANS level	JANTX and JANTXV levels	JAN level
3a	Temperature cycling	Temperature cycling	Temperature cycling in accordance with JANTX level MIL-PRF-19500.
3c 1/	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)	Thermal impedance (see 4.5.3)
9	I_{R1} and V_{FM2}	Not applicable	Not applicable
10 2/	T_A = +90°C; V_{RWM} = 40 V (pk); I_O = 0, half sine wave, f = 60 Hz	T_A = +90°C; V_{RWM} = 40 V (pk); I_O = 0, half sine wave, f = 60 Hz	Not applicable
11	$\Delta I_{R1} \leq$ 100 percent of initial reading or .05 mA, whichever is greater. $\Delta V_{FM2} \leq \pm$ 50 mV dc.	I_{R1} and V_{FM2}	Not applicable
12	See 4.3.2	See 4.3.2, t = 48 hours	Not applicable
13	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .05 mA, whichever is greater; $V_{FM2} \leq \pm$ 50 mV dc.	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .05 mA whichever, is greater; $V_{FM2} \leq \pm$ 50 mV dc.	Not applicable

1/ Thermal impedance shall be performed any time after sealing provided. Temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal impedance.

2/ Junction temperature (T_J) is not to exceed +100°C with V_{RWM} = 40 V (pk). T_J is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, T_J remains approximately equal to T_A or T_L for I_O = 0.

4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-PRF-19500, appendix E. As a minimum, die shall be 100 percent probed in accordance with group A, subgroup 2.

4.3.2 Burn-in conditions. Burn-in conditions are as follows: I_F = 5 A dc min. T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). Mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B.

4.3.2.1 Mounting. Devices may be mounted using any convenient method including the temporary attachment of leads on US suffix devices, provided that the parts are burned-in at T_J = +110°C ± 15°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 appendix E, table V, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current ..... 1 mA to 10 mA.

b. I_H forward heating current ..... 10 A to 20 A.

c. t_H heating time ..... 10 ms.

d. t_{MD} measurement delay time ..... 100 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, tables VIa and VIb (JANS, JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	I_{FSM} = 150 A (pk), condition A 2, I_O = 5 A dc; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
4	1036	I_F = 5.0 A dc; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); $t_{on} = t_{off}$; 3 minutes minimum for 2,000 cycles.
5	1027	T_A = +75°C ± 25°C; I_F = 5.0 A dc (minimum) with I_F adjusted to achieve an average lot. T_J = +125°C +0°C, -10°C.
6	3101	$R_{\theta JL}$ = 20°C/W; L = 9.52 mm (.375 inch) lead length (non-surface mount). $R_{\theta JEC}$ = 10°C/W (surface mount).

4.4.2.2 Group B inspection, appendix E table VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
2	4066	I_{FSM} = 150 A (pk), condition A 2, I_O = 5 A dc; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
3	1027	I_O = 5 A dc; f = 50-60 Hz; V_{RWM} = 40 V (pk). T_L +55°C, lead length = 9.52 mm (.375 inch).
4	2075	In accordance with 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

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4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
2	2036	Tension: Test condition A; weight = 20 pounds; t = 15 s. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for US suffix devices.
6	1026	I_O = 5 A dc; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_{RWM} = 40 V (pk). $T_L \leq \pm 55°C$, lead length = 9.52 mm (.375 inch).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and the conditions for subgroup testing in table II herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in MIL-STD-750.

4.5.2 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.3 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum) prior to shipment. Twenty-two samples shall be serialized and provided to the qualifying activity for test correlation. The maximum upper control limit for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control (SPC). When three successive sealing lots have exhibited control, the data from these three lots will be used to establish a fixed screening limit, (not to exceed the group A, subgroup 2 limit). Once a fixed limit has been established, monitor all future sealing lots using a three-piece sample from each production lot to be plotted on the applicable X, R chart.

4.5.3.1 For initial qualification and requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum) prior to shipment. Twenty-two samples shall be specialized and provided to the qualifying activity for test correlation.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.3	$Z_{\theta JX}$		2.5	°C/W
Forward voltage	4011	I_{FM} = 3.0 A (pk) pulse method (see 4.5.1)	V_{FM1}		0.44	V
	4011	I_{FM} = 5.0 A (pk) pulse method (see 4.5.1)	V_{FM2}		0.47	V
	4011	I_{FM} = 15.7 A (pk) pulse method (see 4.5.1)	V_{FM3}		0.60	V
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM1}		0.20	mA
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM2}		25.0	mA
Forward voltage	4011	I_F = 5.0 A (pk) pulse method (see 4.5.1)	V_{FM4}		0.47	V
Low temperature operation:		T_A = -55°C				
Reverse current leakage	4016	V_{RM} = 40 V (pk) pulse method (see 4.5.1)	I_{RM3}		1.00	mA
Forward voltage	4011	I_F = 5.0 A (pk) pulse method (see 4.5.1)	V_{FM5}		0.62	V

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection 2/	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071	Test condition E	
Electrical measurement		See table I, group A, subgroup 2 herein.	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A; 1,000 hours, see 4.3, screen 10	
Electrical measurement		See table I, group A, subgroup 2 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, (forward voltage drop diode method)	4081 or 3101	$R_{\theta JL}$ = 20°C/W maximum at 9.52 mm (.375 inch) lead length; $R_{\theta JEC}$ = 10°C/W maximum; method 3101 in accordance with 4.5.3 except, I_H = 5 A and T_H = 20 s (minimum)	
Subgroup 5			
Not applicable			
Subgroup 6			
Not applicable			

1/ For initial design and process change verification only (one time testing).
2/ For sampling plan, see MIL-PRF-19500.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5822) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	55801	- - -
1N6702	JANHCA1N6702	- - -
	JANKCA1N6702	- - -

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1851)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/629	**2. DOCUMENT DATE (YYMMDD)** 970612

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, RECTIFIER, SCHOTTKY BARRIER, TYPES 1N5822 AND 1N5822US JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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